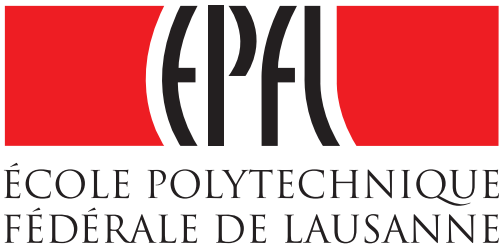
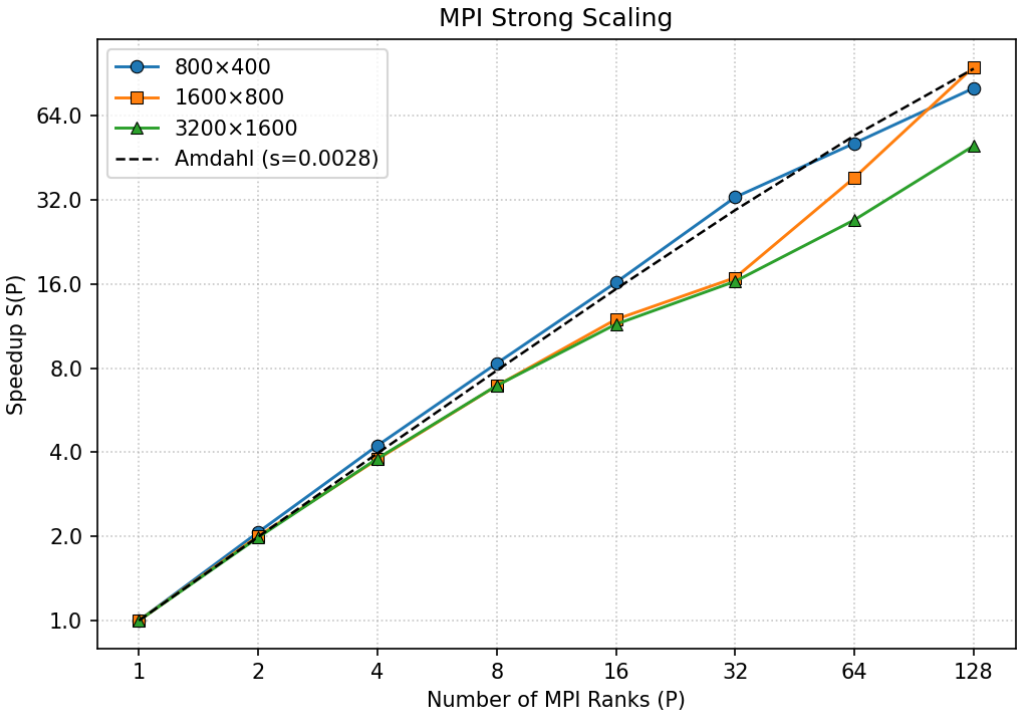
# Parallelisation of Lattice Boltzmann Solver
# MATH-454: Parallel and High Performance Computing

Ahmed Rockey Saikia (416140)
École Polytechnique Fédérale de Lausanne, Switzerland

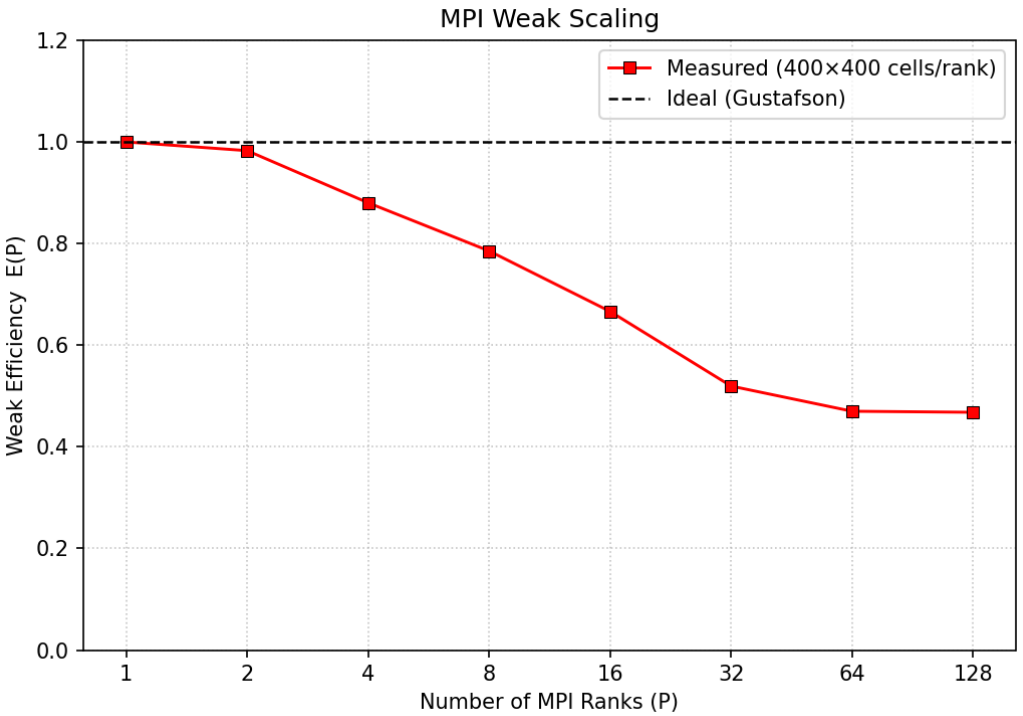


# MPI Strong Scaling




- Three grid sizes with $P = 1$ to 128 (two nodes at $P = 128$).
- Bandwidth-bound: efficiency falls as ranks compete for bandwidth.
- $800 \times 400$: Auxiliary to compute cell ratio $\eta = 2P/n_x \to 32\%$ at $P = 128$ implies communication bound.
- $1600 \times 800$: $S(128) = 95.8$. Measured against Amdahl ($s = 0.28\%$, $S_{\text{Amdahl}}(128) \approx 94$).

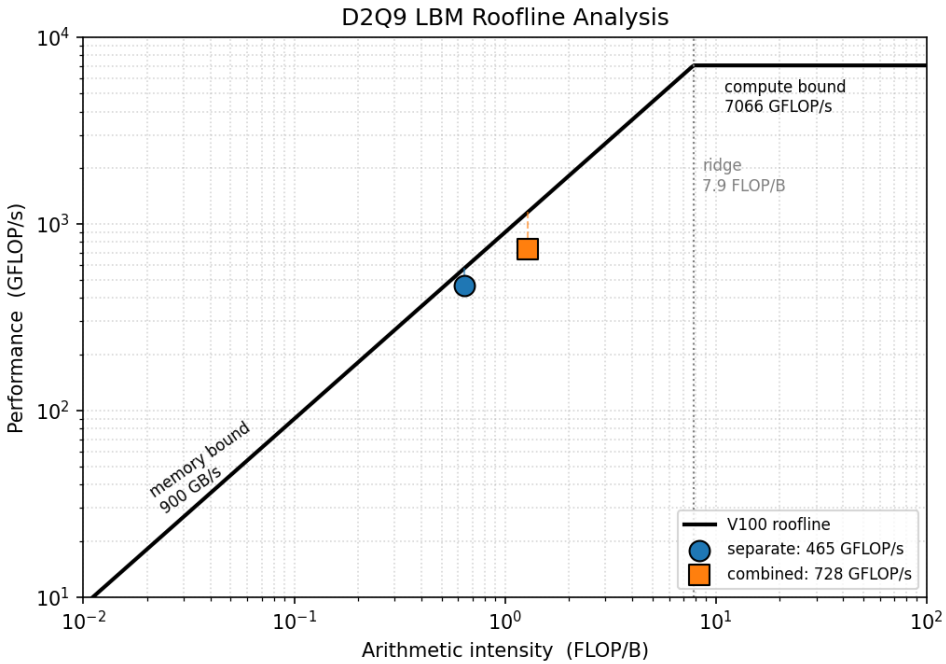
# MPI Weak Scaling




- Fixed $400 \times 400$ K cells/rank; Gustafson ideal $E = 1$.
- $E$ falls $100\% \rightarrow 47\%$ at $P = 64$ (memory bandwidth).
- Efficiency drop relaxes at 32 to 64 reflecting the two CPU sockets.
- $64 \rightarrow 128$ ($1 \rightarrow 2$ nodes): wall time constant $\sim 25$ s, MLUPS increases $793 \rightarrow 1580$ ($\times 2$).

$\Rightarrow$ Per socket bandwidth contention.

# CUDA Performance




- One thread per cell, SoA layout.
- Kernels: *separate* 2513 MLUPS & *combined* 3937 MLUPS but reduced achieved bandwidth (567 GB/s).
- Memory-bound: $\mathrm{AI} = 0.64(\textit{separate})$ & $1.28(\textit{combined}) \ll$ ridge 7.85 FLOP/B.
- `block_x` $\geq 32$ for coalescing: $1913 \rightarrow 2558$ MLUPS.
- SoA $\sim 4\times$ AoS; achieves $\sim 724$ GB/s.

# Anomaly: Strong-Scaling Efficiency

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

| $P$ | Data/rank | $S(P)$ | $E(P)$ |
|---|---|---|---|
| 32 | 2.88 MB | 16.9 | 52.7% |
| 64 | 1.44 MB | 38.5 | 60.1% |
| 128 | 0.72 MB | 95.8 | 74.9% |

$1600 \times 800$ grid
shared L3 size = 54 MB (JED);
# of cores per socket = 36
L3 size per core = 1.5 MB,
$P^\star = \frac{92.2\text{MB}}{1.5\text{MB}} \approx 61$

- Efficiency rise after $P = 32$ showing super-linear increment $\times 2.28$, $\times 2.49$.
- $S(128) = 95.8 >$ Amdahl 94.4, cache effect outside Amdahl's model.
- Cause: data per rank $92.2/P$ MB crosses the L3 threshold $P^\star \approx 61$ and fits in cache.
- $3200 \times 1600$ ($P^\star \approx 246$) never crosses it therefore efficiency declines monotonically to 39%.